1. Representation leads for binary point decomposition

Campaign `research/representation_leads_20261109/`, 2026-11-09. Exact derivations in native Rust (`derive/`), no timings, no experiment runs. **Every number is an unverified assertion:** no pcc claim was minted, and the `research-kb` server was unreachable, so novelty rests on the lead atlas `--match` and one literature search.

1.1. Why these representations

An atlas finding on ECC2K-130 (KN-FIND-aa2efc, unreviewed) gives the product law relations × targets × attempts = m·2^{131}. With free attempts the floor is $2^{68.58}$ at m = 3 (above rho's $2^{60.81}$), $2^{56.40}$ at m = 4, $2^{48.44}$ at m = 5 and $2^{42.85}$ at m = 6. A representation therefore matters only if it makes one attempt fit roughly 2^{12} (m = 5) to 2^{18} (m = 6) rho-step equivalents: the leads aim at m ≥ 5.

1.2. R1′: quadratised chains (line norm form)

For a merge $P_a + P_b = U$ on $y^2 + xy = x^3 + ax^2 + b$, the line through P_a, P_b and $-U$ gives $e_1 = \lambda^2 + \lambda + a$, $e_2 = \mu$, $e_3 = \mu^2 + b$ (unit-tested on 1000 random pairs). Keeping $\mu = e_2$ replaces the cubic $S_3(x_a, x_b, x_U)$ by two quadratics: $x_a x_b + x_U(x_a + x_b) = \mu$ and $\mu x_U + x_U^2(x_a + x_b) + \mu^2 + b = 0$. The descended chain then has degree 2 at every m.

Size (exact). Against the confound control (w = $x_a x_b$ on the same merges), $\mu = e_2$ has the same variable count and the same set of distinct monomials, and 30–40 % more terms (figure below). On written size it is a relabelled generic degree reduction.

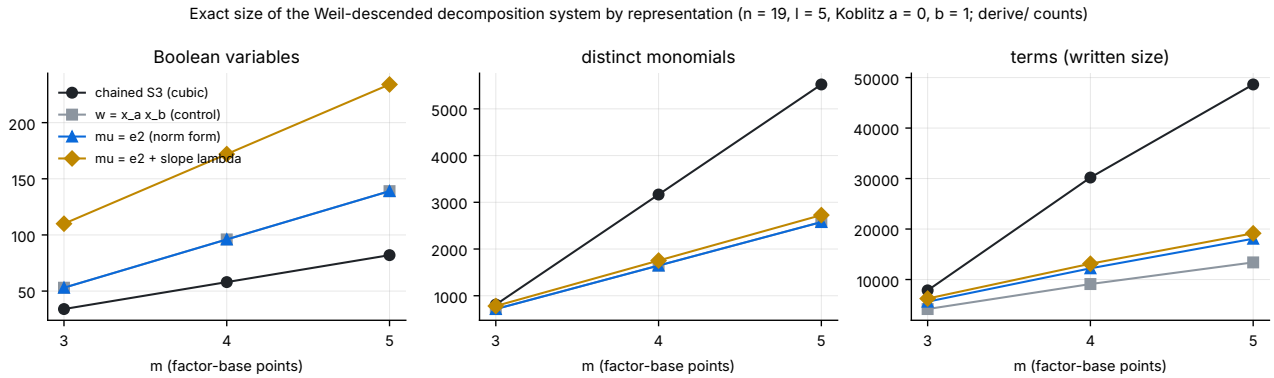


Figure 1: Exact sizes at n = 19, l = 5 (Koblitz a = 0, b = 1). All 288 built systems satisfy their planted relation; single-bit flips are detected in ≥ 31/32 cases per system.

The slope λ. Over GF(2^n), $\lambda^2 + \lambda = c$ is solvable exactly when Tr(c) = 0. An exhaustive census of every root triple of $S_3(x_1, x_2, X) = 0$ (n = 11, 13; four curves each) finds only two classes: all three abscissae on E with λ in the field, or all three on the twist with none. Mixed triples never occur, and every sampled on-E triple lifts to a true sum. In a chain the merges share abscissae and the last contains x_R on E, so the trace equations are **implied**; the atlas already records the trace linear equation (KR-IC-f5c584). λ is eliminable (one linear equation per merge).

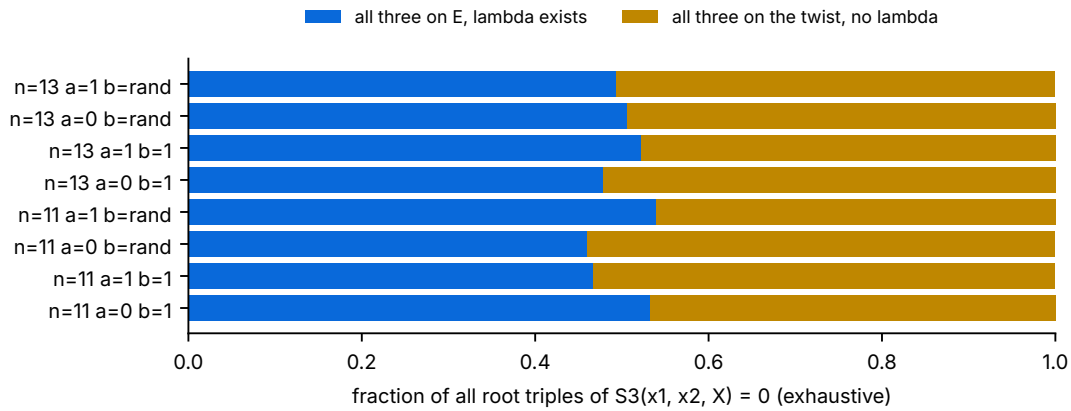


Figure 2: Exhaustive census of S_3 root triples (results/census.jsonl).

Open question (gated). Whether $\mu = e_2$, or the explicit implied trace equations, lowers the refutation degree relative to the generic control. `PROTOCOL_DRAFT.md` gives the four-arm cell; it is not frozen (instrument binding missing) and cannot run without the checkpoint token.

1.3. R3: odd power-sum coordinates

In characteristic 2, $p_k = \sum_i x_i^k$ is slot-separable on a subspace. Exact linear algebra over GF(2^{13}) and GF(2^{17}), three (b, x_R) draws each: $p_1 S_3$ lies in $F[p_1, p_3, p_5, \ldots]$, which is the generic pair-table form; $p_1^j S_4$ does not, for any j ≤ 8, on all six draws. S_4 is in the elementary-symmetric span, and degree-12 positive controls in three variables are recovered (j = 0), so the method sees such elements. Closed in scope.

1.4. R2: membership as a Frobenius chain modulo N

$N = \prod(X - x_i)$ divides the subspace polynomial L_V iff all roots lie in V; tracking $R_k = X^{2^k} \bmod N$ with $R_k = R_{k-1}^2 \bmod N$ is bilinear, so membership is degree 2 for all m at l·m·n auxiliary bits. This is very likely one of the big-field systems analysed by Huang–Kosters–Yeo (CRYPTO 2015, "Last fall degree, HFE, and Weil descent attacks on ECDLP"); the paper was not retrieved, so the lead is blocked on that reading rather than closed.

1.5. R4 and R5

R4 (uncorrelated slots for the n = 83, m = 10 gate) is cell G6 of PR #2679 and waits for the checkpoint token. R5 (mixed chains) is the R1′ construction itself: auxiliaries only on merges whose S_3 is cubic.

1.6. Scope and limits

- Sizes are at toy n (11–19); no solving degree, timing or cost is measured.
- Census and power-sum results are exact on the stated fields and draws only.
- Nothing here changes a leaderboard, benchmark or validation row.